Towards Closed-loop Deep Brain Stimulation via Optimal Control

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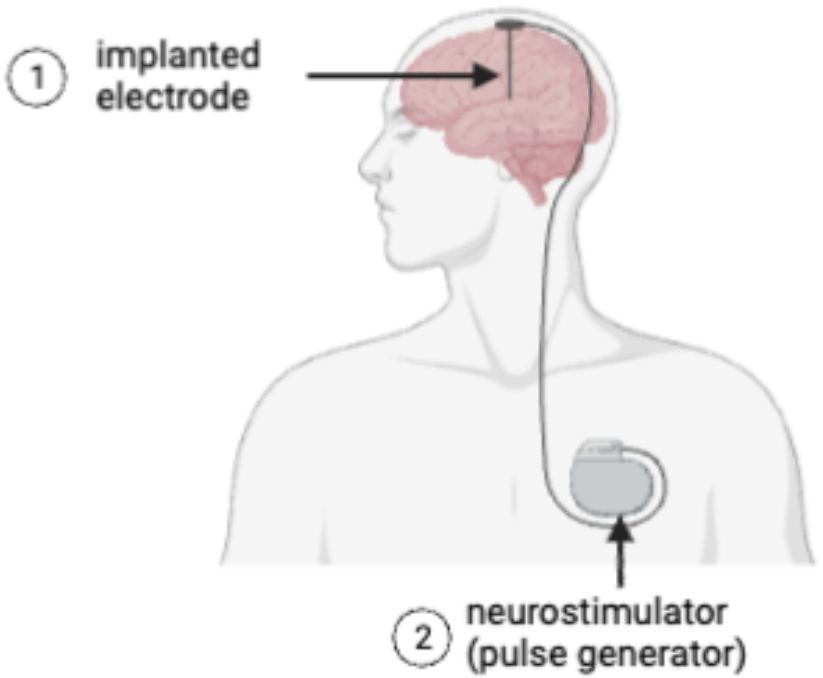
EMORY UNIVERSITY

Hitchhiker's Guide to DBS

Deep Brain Stimulation (DBS)

- Delivers electrical pulses to the brain via surgically implanted electrodes.
- Used for medication resistant neurological disorders e.g., Parkinson's Disease.



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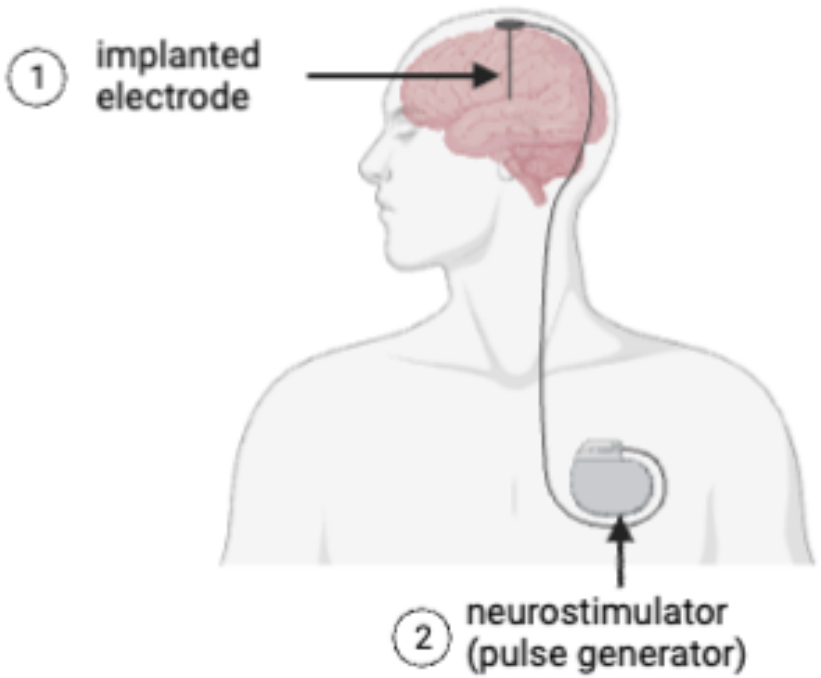
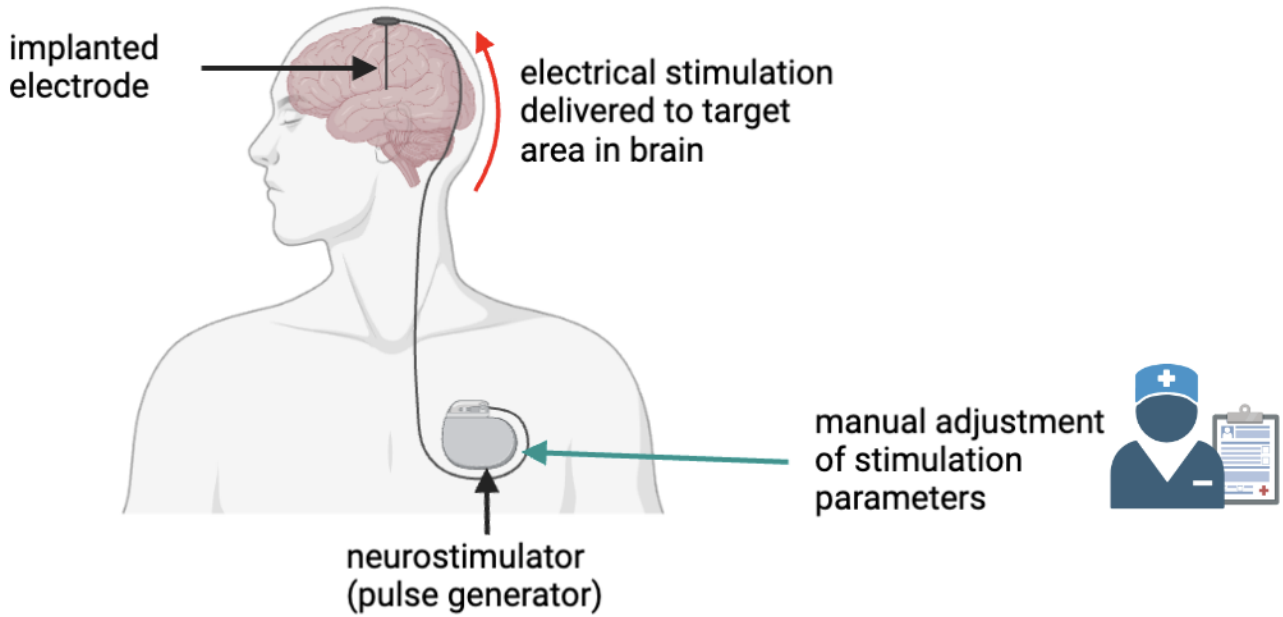


Clinical Goal: **Find optimal stimulation to manage symptoms**

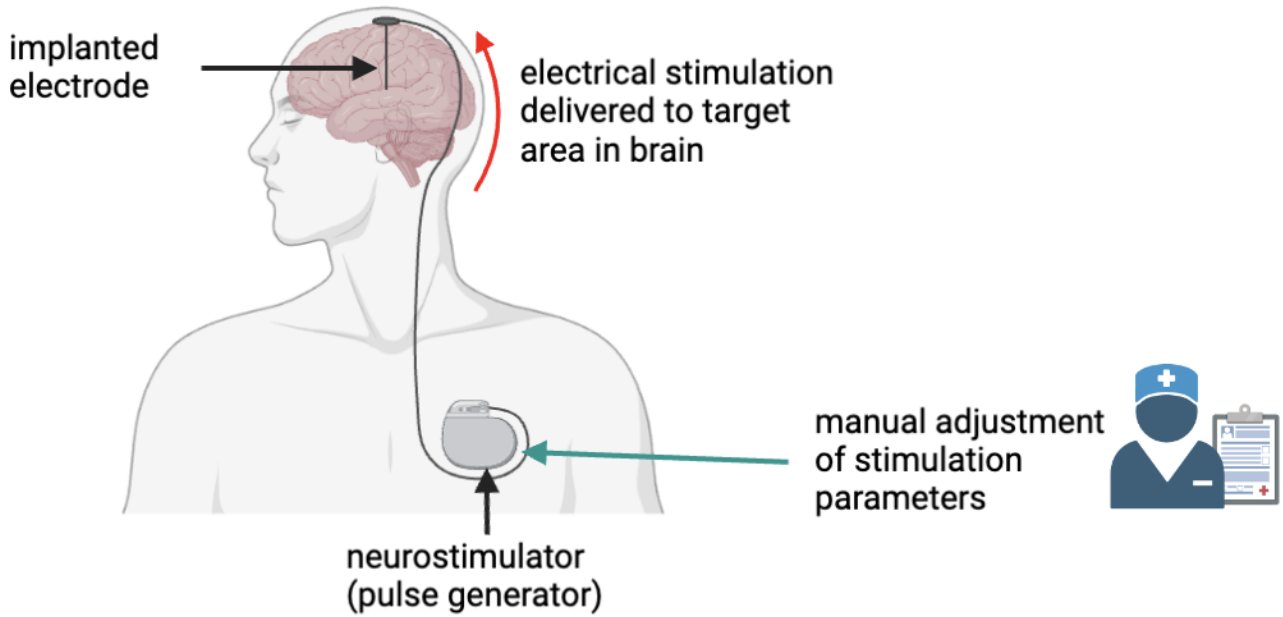
Open-loop DBS



Overview

1. Clinicians **manually program** the neurostimulator (often via trial-and-error)

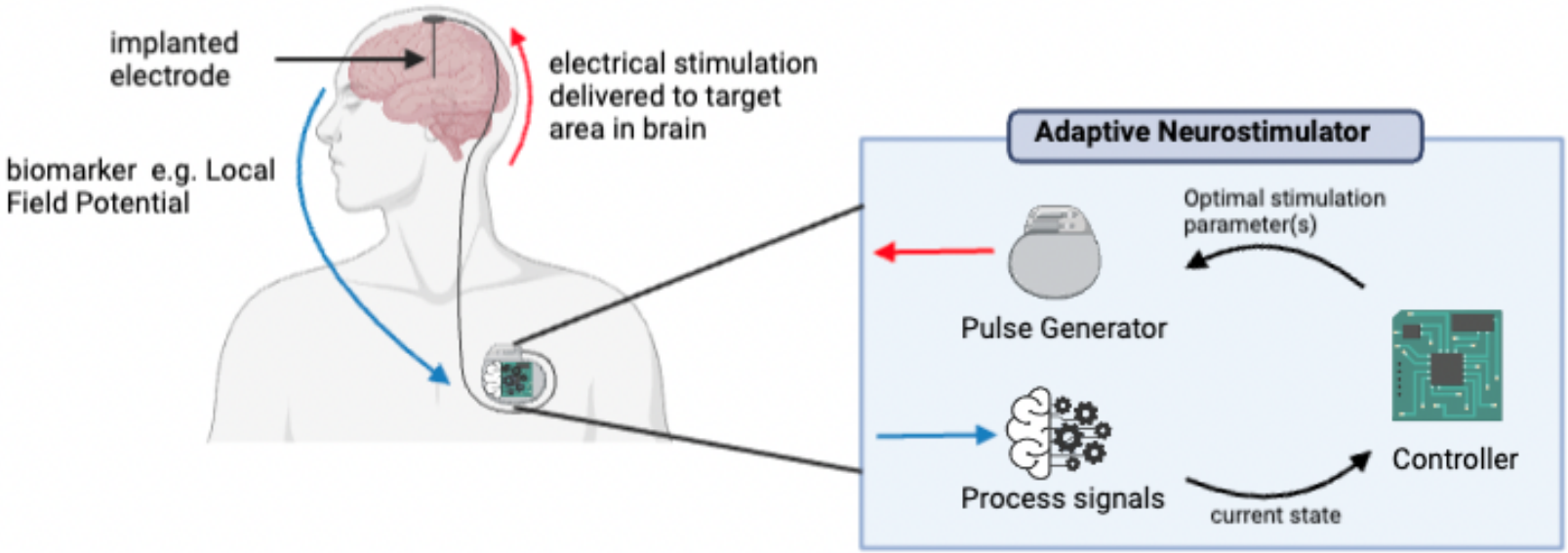
Open-loop DBS



Overview

1. Clinicians **manually program** the neurostimulator (often via trial-and-error)
2. **Continuously** deliver a **fixed** amount of electrical impulse to patient.

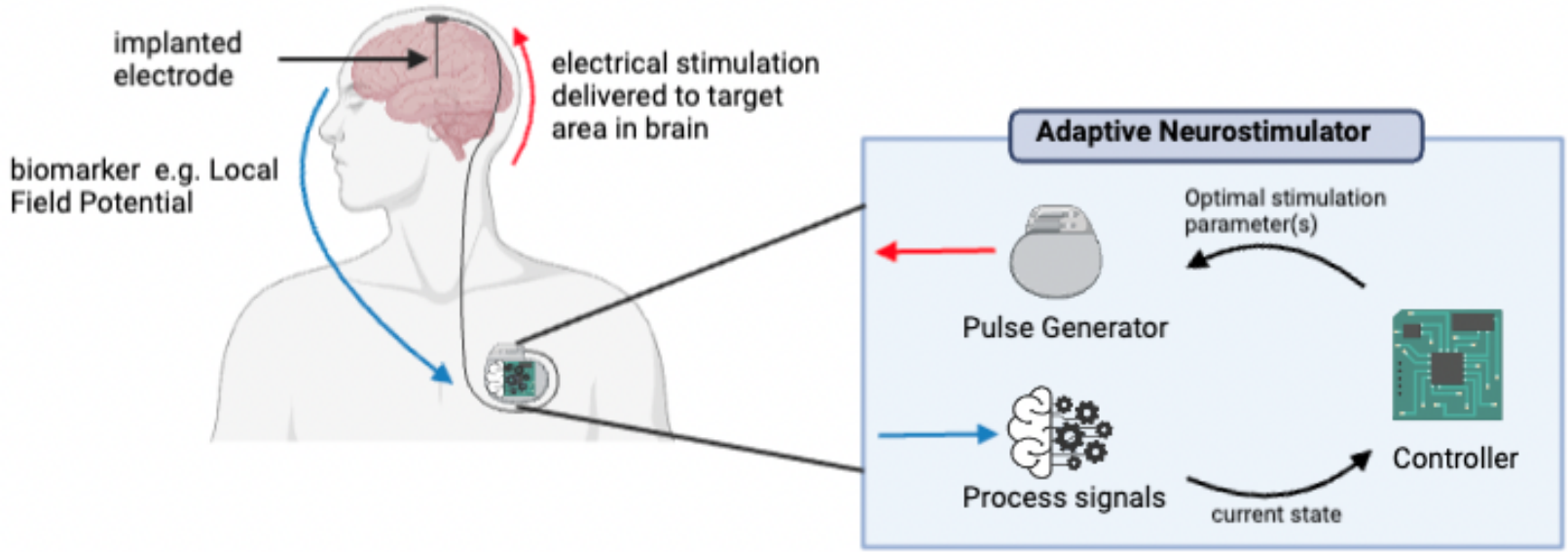
Closed-loop DBS



Overview

1. Controller adjusts the stimulation based on **feedback**

Closed-loop DBS



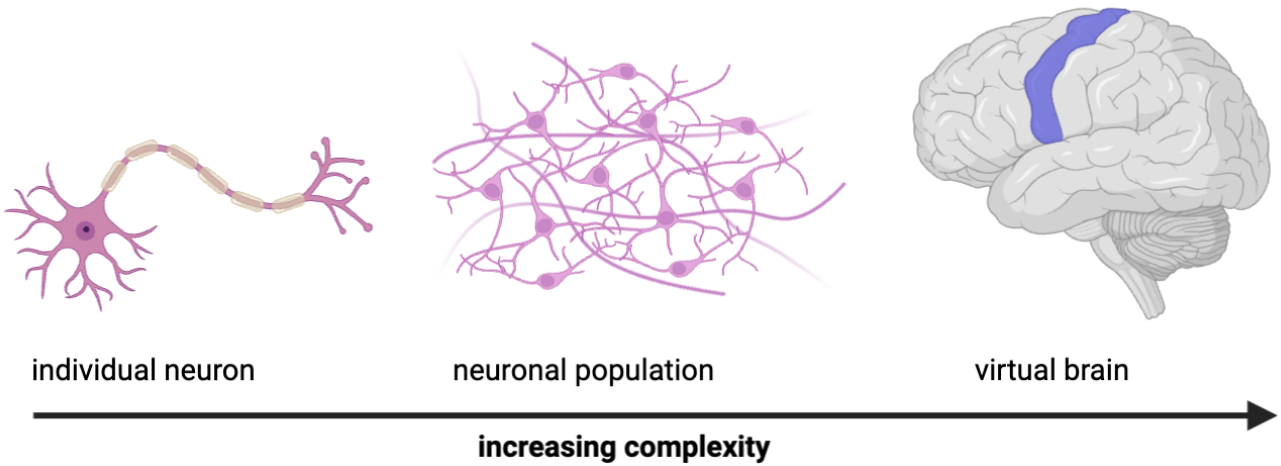
Overview

1. Controller adjusts the stimulation based on **feedback**
2. **Adaptive**: only deliver electrical stimulation when needed

DBS Meets Optimal Control

A rich application area for Deep learning and Optimal Control. Possible research directions:

1. Modeling dynamics for a single neuron to large neuronal populations
2. Characterizing neurostimulation as a control problem
3. Applying control-based approaches to solve the problem
    - Combine Deep Learning and Optimal Control



Control Formulation

Consider neuronal dynamics given by

$$\frac{dz}{dt}(t) = f(t, z(t)) + e_1 u(t),\ t \in [0, T],\ e_1 = [1, 0, 0, 0]^\top,$$

with initial data $z(0) = x$ where

- $z(t) \in \mathbb{R}^4$ denotes the **state** variables (i.e., membrane potential z_0 and ion channel gating variables: z_1, z_2, z_3)
- $u(t)$ is the **control** variable (external current provided as input at time t).
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$$f(t, z(t)) = \begin{pmatrix} -\big(I_{\mathrm{Na}}\big(t, z_0(t)\big) + I_{\mathrm{K}}\big(t, z_0(t)\big) + I_{\mathrm{L}}\big(t, z_0(t)\big)\big) \\ \alpha_m(z_0(t))(1 - z_1(t)) - \beta_m(z_0(t))z_1(t) \\ \alpha_n(z_0(t))(1 - z_2(t)) - \beta_n(z_0(t))z_2(t) \\ \alpha_h(z_0(t))(1 - z_3(t)) - \beta_h(z_0(t))z_3(t) \end{pmatrix}.$$

where

- $I_{\mathrm{Na}}, I_{\mathrm{K}}, I_{\mathrm{L}}$ are sodium, potassium, and leak ion channel currents
- α_x and β_x are voltage-dependent rate variables with $x \in \{m, n, h\}$

Optimal Control Problem

Objective

Find an optimal control $u^*(t)$, $0 \leq t \leq T$ that **minimizes** the **value function**:

$$\Phi(t, z(t)) = \inf_u \left(\int_t^T L(s, z(s), u(s)) ds + G(z(T)) \right),$$

where

- $L(t, z, u)$ is the running cost (Lagrangian cost)
- $G(z(T))$ is the terminal cost

Methods for finding optimal solution:

1. local: generates a control policy based on a fixed initial state $z(0)$.
2. global: find OC policy for any given state $z(t)$, $t \in [0, T]$

Local Solution Method: PMP

- Pontryagin's Maximum Principle provides necessary conditions for $u^*(t)$
- **Optimality conditions** for $t < s \leq T$ are given by

$$\begin{cases} \partial_t p(s) = \nabla_z \mathcal{H}\big(s, z^*(s), p(s), u^*(s)\big), \\ \partial_s z^*(s) = -\nabla_p \mathcal{H}\big(s, z^*(s), p(s), u^*(s)\big) \\ z^*(0) = z,\ p(T) = \nabla_z G(z^*(T)) \end{cases}$$

where

- $\mathcal{H}$ is the Hamiltonian of the system, defined as

$$\mathcal{H}(t, z, p, u) = -L\big(t, z(t), u(t)\big) - p^\top \cdot [f(z(t), t) + e_1 u(t)]$$

- p is the costate/adjoint variable.

Then $u^*(s) \in \arg\max_u \mathcal{H}\big(s, z^*(s), p(s), u(s)\big)$

Global Solution Method: HJB

- Provides necessary and sufficient conditions for optimality.
- At optimality, Φ **satisfies the HJB equation**:

$$\begin{cases} -\partial_t \Phi(t, z(t)) + \sup_{u \in U} \mathcal{H}\big(t, z, \nabla_z \Phi(t, z(t)), u(t)\big) = 0, \\ \Phi(T, z(T)) = G(z(T)), \end{cases}$$

- The optimal control u^* can be recovered online in **feedback form** as

$$u^*(s) \in \arg\max_u \mathcal{H}\big(s, z^*(s), \nabla_z \Phi(s, z^*(s)), u(s)\big)$$

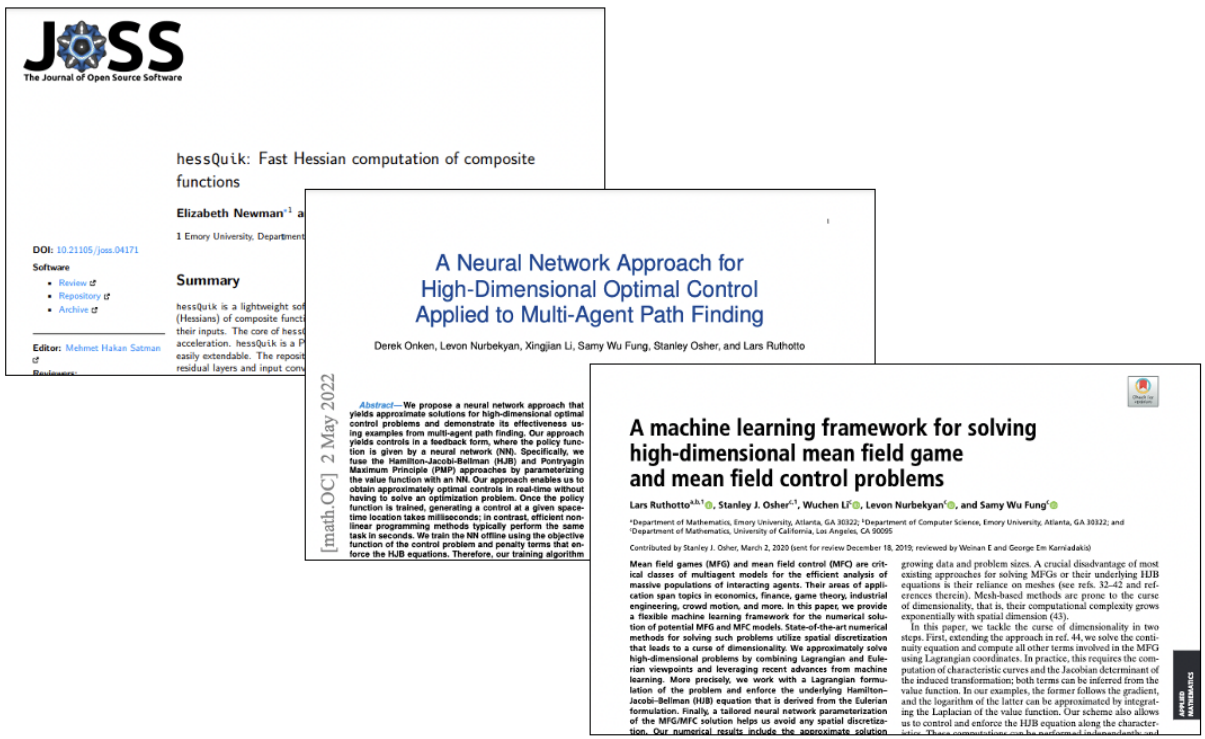
Note: In HJB, optimal controls depend on $\nabla_z \Phi$, i.e.,

$$p(s) = \nabla_z \Phi(s, z(s)), \quad t < s \leq T.$$

A semi-global approach

NeuralHJB

- Combines PMP and HJB approaches while parameterizing Φ with a neural network as per Ruthotto et al. [2020] and Onken et al. [2021].
- **Today @ 3:15 PM in Palm Room 6** Advances and Challenges in Solving HJB Equations Arising in Optimal Control. Deepanshu Verma [MS65 Part II]



Learning Problem

Approximate Φ using a neural network parameterized by θ, denoted by Φ_θ

$$\Phi_\theta(x) = w^\top N(x;\theta) + \frac{1}{2}x^\top(A^\top A)x + b^\top x + c$$

for space-time inputs $x = (t, z)$, trainable weights $\theta = (w, A, b, c)$, and a residual neural network $N(x;\theta)$.

Objective

$$\min_\theta \mathbb{E}_z \mathbb{E}_{x\sim\rho}\left\{\ell(T) + g(z(T)) + \beta_1 c_{\mathsf{HJ}}(T) + \beta_2 |\Phi_\theta(T, z(T)) - g(z(T))|\right\}$$

subject to

$$\partial_s \begin{pmatrix} z \\ \ell(s) \\ c_{\mathsf{HJ}}(s) \end{pmatrix} = \begin{pmatrix} -\nabla_p H\big(s, z, \nabla_z \Phi\big(s, z\big)\big) \\ L(s, z, u^*) \\ \big| - \partial_t \Phi\big(s, z(s)\big) + H\big(s, z, \nabla_z \Phi_\theta\big(s, z, u^*\big)\big)\big| \end{pmatrix},$$

for $z(0) = x$, $\ell(0) = 0$, and $c_{\mathsf{HJ}}(0) = 0$,

where β_1 and β_2 are hyperparameters, c_{HJ} is a penalty term derived from HJB

An Example: Target Potential $-20mV$

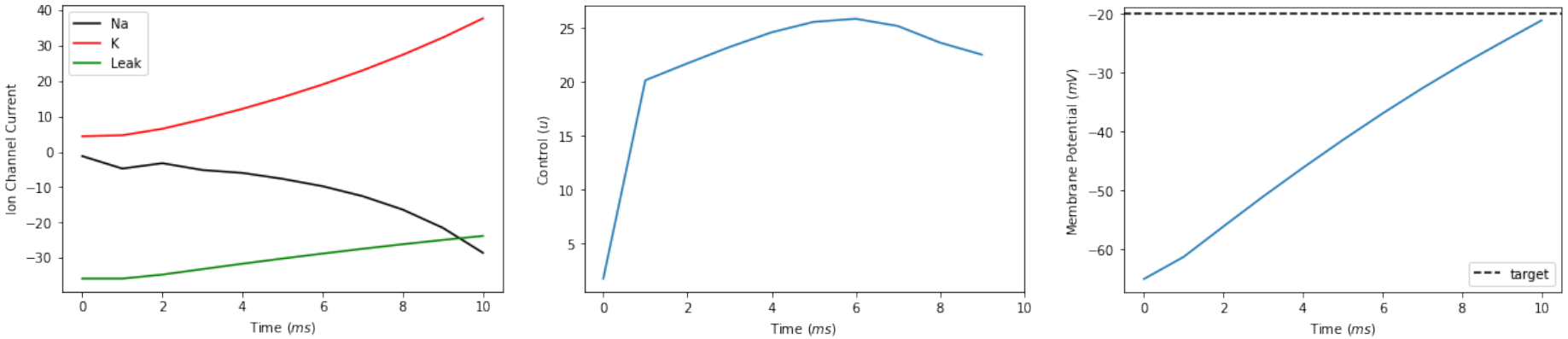


Figure: Observed ion channel currents, controls, and membrane potential from a local solution method

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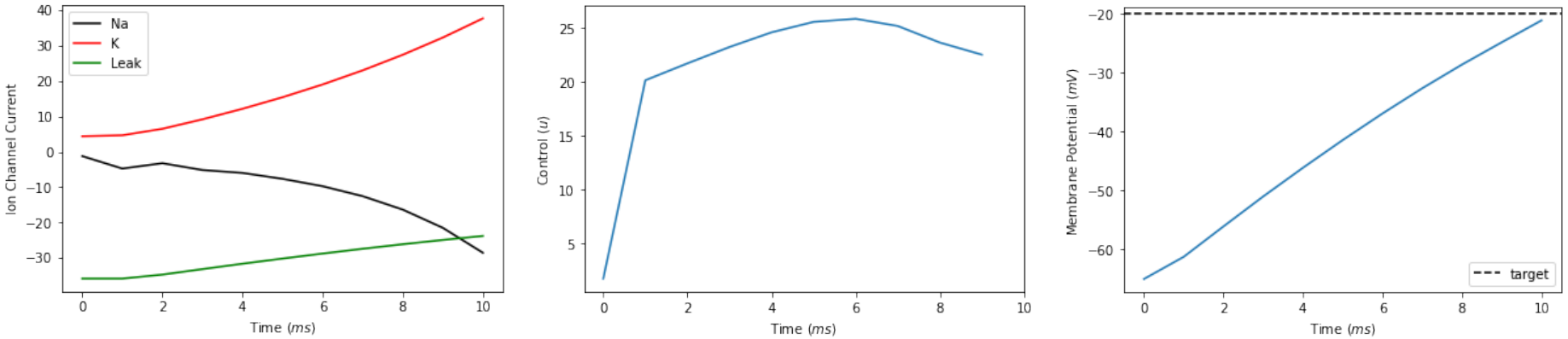

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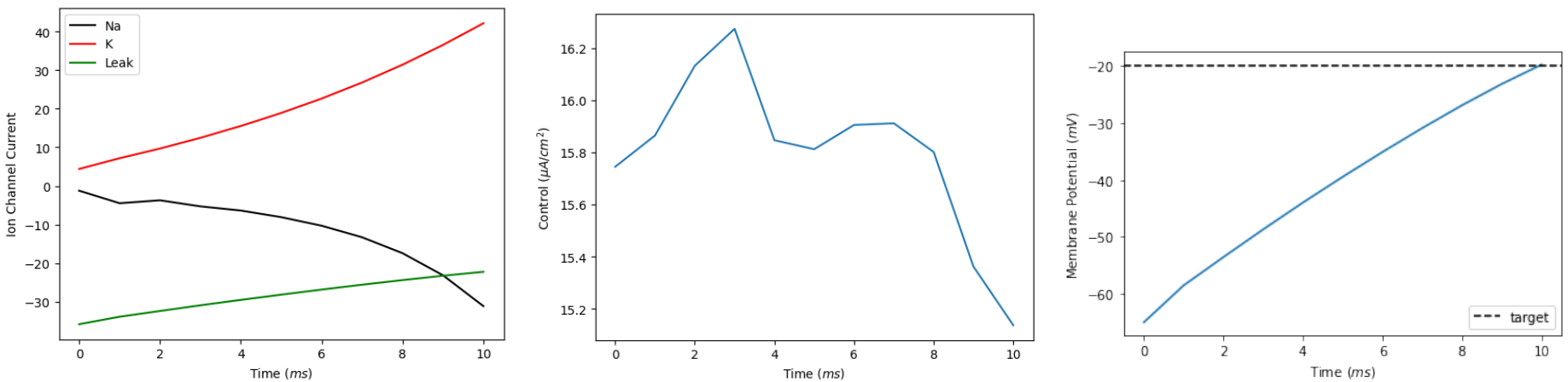

Figure: Observed ion channel currents, controls, and membrane potential from semi-global solution method

Promising Research Directions

Beyond Single Neuron Dynamics

- Extension to large neuronal populations and network models
- Neural mass models approximate large-scale dynamics
- Opportunities to apply mean field theory and methods of finding approximate solutions to high-dimensional problems

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Reinforcement Learning for DBS

- OC is generally model-based and solving OC problems can be challenging
- RL is ideal when model is not available/accurate
- Opportunities to build virtual environments, cast control problem as a learning problem, and apply RL machinery

Summary

Some takeaways:

- Demonstrated the possibility of extending DL+OC to DBS
- Conventional (open-loop) DBS approaches can be improved
- Closed-loop DBS has high practical importance: for neurotech and patients
- Plenty of challenges and opportunities for researchers

Thank You!

Questions, comments, suggestions, collaborate?

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